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Initially concealed concussion lowers in-game performance of NCAA Division I football players: A case series

Aaron J. Monseau, MD[1], Brenden J. Balcik, MD[1], Luke Roberts[2], Reed Andrews[2], Melinda J. Sharon, MPH[3]

[1]*Department of Emergency Medicine, School of Medicine, West Virginia University*

[2]*School of Medicine, West Virginia University*

[3]*School of Medicine, Marshall University*

Corresponding Author:

Dr. Aaron J. Monseau

1 Medical Center Drive

Morgantown, WV, 26506

Email: amonseau@hsc.wvu.edu

Phone: 304-293-2436

Fax: 304-293-6702

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ABSTRACT

Objectives: Many athletes conceal concussion symptoms to continue to play. If athletes and coaches knew that performance drops significantly after a concussion, in turn hurting the team, athletes may be more prone to report concussion symptoms. The objective of our research was to report three cases that support this performance decline after an undisclosed concussion. **Methods:** This is a case series of three NCAA Division I football players who did not immediately report their concussions. The play in which the concussion occurred was identified via video review then the players' pre and post-injury performance grades, as scored by their position coaches, were recorded. Additional data regarding their performance in other games and their concussion assessments were also recorded. **Results:** Three injuries were identified and reviewed. Player 1 had a pre-injury performance score of 76/100 with a post-injury score of 47/100. This was a drop of 29 points for a decrease in his in-game performance by 38%. Player 2 had a pre-injury score of 100/100 with a post-injury score of 60/100. This was a drop of 40 points or a 40% decrease in his in-game performance. Player 3 had a pre-injury score of 67/100 and post-injury score of 43/100 for a drop of 24 points or a 36% decrease in his in-game performance. **Conclusions:** We reported three college football players who suffered concussions but did not immediately disclose their symptoms. Each had a marked decrease in performance grades after their concussion. If the performance effect of a concussion can be quantified, athletes and coaches may begin to change their mindset that a concussion can be "played

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through." This case series should serve as a first step in quantifying concussion effects on performance.

Keywords: *Concussion, Effects, Football, Reporting, Disclosing*

INTRODUCTION

Sport-related concussions (SRC) have become a major point of research and public policy discussion over the last decade. Injury data for the five year period from the 2009-2010 to 2013-2014 school-years estimates that 10,560 sport-related concussions occur every year in NCAA athletes with 3,417 of those in football, equating to 6.71 per 10,000 athlete-exposures [1]. These numbers likely under-represent the actual number of concussions, as athletes continue to hide their symptoms and refuse to report their injury [2-4].

The medical community instructs athletes to disclose concussions by warning that, while the exact risk is still debated, competing with an active concussion may result in prolonged symptoms, disqualification from an extended period of competition, acute cerebral edema, intracranial hemorrhage, or even death [5,6]. The short-term effects of concussion such as headache, dizziness, mental fogginess, photophobia, balance problems, cognitive processing issues, and visual acuity decline have been well documented [7-9]. In addition to the intracranial sequelae of head injuries, there is some evidence of an increased risk of lower extremity injury such as anterior cruciate ligament tear and ankle sprain that has been demonstrated in multiple studies which has been hypothesized as the result of deficits in postural control or perception-action coupling [10-15].

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Despite these warnings and evidence that immediate removal from activity improves outcomes, athletes continue to compete after a concussion [16]. Coaches and athletes do not fully respect the serious consequences that could come from continued participation while suffering from a concussion [2]. Torres et al. conducted an anonymous survey of collegiate athletes, where 39 out of 70 (56%) continued to play after experiencing concussion symptoms, and 30 out of 70 (43%) admitted to hiding a concussion to stay in a game [17]. Another study of 161 college athletes at graduation revealed that 19 (11.8%) had experienced an unreported concussion, with 8 of the 19 (42.1%) citing "*Did not want to let down teammates*" as a reason for not reporting [18]. There may be some evidence that reporting rates are improving slightly, but the current numbers of known concussions may still be the tip of the iceberg [19-25].

There is a growing body of literature attempting to assess performance after recovery from a concussion in several different ways across several different sports. Zuckerman et al found that the majority of National Football League players showed no deficit in performance when gameplay statistics were compared between the three games before a SRC and the three games after returning from a SRC when using a control group of players with other types of injuries [26]. A similar study in National Basketball Association players also found no deficit in performance in the first five games after returning from a SRC [27]. In a series of studies in National Hockey League players with SRC versus other injuries, no difference was found upon return from SRC in measures of game performance in both typical and more advanced statistical metrics [28-30].

While these studies have not shown a decrease in performance after recovery from a concussion, we still believe that there is a temporary decrease in performance in the acute period after a SRC even when undetectable to medical personnel. If it were known that athletic

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performance will significantly decline immediately after a concussion, athletes may be more likely to promptly disclose their injury. They may begin to see that hiding their injury and continuing to play may actually be more detrimental to their team than disclosing their injury and coming out of the game or practice. We hypothesized that in-game performance would be negatively affected after a concussion so we are reporting three cases of NCAA Division 1 football players who suffered concussions, did not disclose their symptoms immediately, and had the predicted decline in their in-game performance.

MATERIALS & METHODS

This is a case series of three NCAA Division I Football Bowl Subdivision football players who did not report their concussions until a later time. The play in which the concussion occurred was identified via video recordings of games. The primary outcome data reported is the players' performance grade (PG) both before and after the concussion. Further description of performance grades can be found in the next paragraph. In addition to the raw scores, we also reviewed their change in PG as a ratio of their pre-concussion PG in an attempt to control for external factors that were unique to the game such as weather, game scheme and, most importantly, quality of opponent which could all affect the score. To prevent bias, we approached coaches for PG only after all scoring was completed.

In the upper levels of American football, the players are routinely graded on their performance each game. The "position coach" for each player generates the PG in the 24 hours after the game. Each position on the field is coached by a "position coach" in addition to the Head Coach and either the Offensive or Defensive Coordinator. After each game, the position coaches score each player in their position group on a set of performance criteria, including being in the correct place before the play, getting to the correct place during the play, and

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executing the correct action during the play. To better understand this scoring, the study team interviewed several coaches. For each play, most coaches mark either a "1" if the majority of the player's actions were correct or a "0" if the majority of the player's actions were incorrect. A player may also receive a "1" for a very good action such as an interception or may receive a "0" for a very bad action such as a fumble. The sum of these individual play scores is then divided by the total number of plays in which they participated. This creates a 0-100% scale where a PG of 100 is perfect performance and a 0 would signify that the player made no correct plays during the game. Generally, a raw PG score over 90 indicates a very good player and below 75 is viewed as a poor player who will only be put in the game if necessary. Most scores are in the 75-90 range.

Secondary data recorded were the components of the Standardized Concussion Assessment Tool-3 (SCAT-3), as that was the most current SCAT available at the time, which includes the number of symptoms, a symptom score (total of symptom ratings from 0-6), the Standardized Assessment of Concussion (SAC), balance score, and Maddock's score. Taking only 7-10 minutes to complete, the SCAT-3 is useful for a comparison with the baseline test which is done before the season starts so the baseline scores were recorded as well. The SCAT-3 was performed both before the season and post-injury by the Athletic Trainer with the team which is encouraged to promote consistency. Normative data for the SCAT-3 in non-concussed college athletes has shown an average of 1.75 symptoms endorsed with average SAC and balance scores of 27.17 (standard deviation=2.01) and 25.64 (SD=4.07), respectively.[31] We also recorded each player's total PG from the other games in which they participated during the same season.

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RESULTS

Case 1

Player 1 was a defensive lineman who suffered a concussion toward the end of the third quarter, when he spun the ball carrier to the ground and struck the back of his head on the playing surface. He did not come out of the game, and no injury was noted by anyone on the sideline. Shortly after the game, he disclosed his injury in the locker room and admitted he was trying to hide his symptoms on the field.

He had played in 38 plays prior to the concussion, with a performance grade of 76/100. For the 17 plays after the concussion, his performance grade dropped 29 points to 47/100, a 38% drop off his initial score. See Figure 1 for graphical representation of the pre-concussion and post-concussion grades. During the season, his average grade was 75/100 in the 11 other games, with a range from 63-82. His position coach noted that he would occasionally have a decrease in performance due to fatigue, but he never dropped this far.

When he disclosed his injury after the game, he had a normal exam. By reviewing the game film after the injury was reported, it appeared that he may have lost consciousness for less than a second. When he completed his post-concussion evaluation including a SCAT-3 test the day after injury, he had 18 of 22 symptoms (baseline was 6 of 22), with a severity score of 64/132 (baseline 12/132). His SAC score was 26/30 (baseline 28/30), a balance score of 7/30 (baseline 5/30), and a Maddock's Score of 5/6. Breakdown of these scores can be found in Table 2. He returned to normal within the week and was cleared for full return to play about 10 days after the concussion.

<Insert Table 1 here>.

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Case 2

Player 2 was a defensive skill player who suffered a concussion toward the end of the first quarter. When tackling a wide receiver, they collided helmet-to-helmet. He complained of neck pain only and came out of the game for two plays. His position coach came to the medical staff to say the player did not "seem right" after he had returned to the game for five more plays. He was evaluated on the sideline and continued to deny concussion symptoms. The only abnormality on sideline exam was diplopia on convergence around 15 centimeters from his nose so he was removed from the game. For the remainder of the game, he continued to deny symptoms and ask to be permitted to return to the game. When examined the following day, he admitted to concealing concussion symptoms at the game because he wanted to be cleared to return.

He had played in 19 plays prior to the concussion, with a performance grade of 100/100. In his five plays after the concussion, his grade dropped 40 points to 60/100, a 40% change from his baseline (Figure 1). During that season, he averaged 89/100, with a range of 80-96 without any noticeable in-game decline. When his position coach reviewed these plays at the authors' request and with the benefit of hind-sight, he noted that the player was not himself for those five plays, and looked out-of-sorts and off-balance.

When completing his post-injury SCAT-3 test one day after the game, he recorded 1 of 18 symptoms (baseline 1 of 18) with a severity of 2/132 (baseline 4/132). His SAC Score was 25/30 (baseline 26/30), with a Balance Score of 2/30 (baseline 5/30), and a Maddock's Score of 5/6 (Table 2). He was back to normal after one day and was cleared for full return to play six days after the injury.

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<Insert Figure 1 here>.

Case 3

Player 3 was an offensive skill player who suffered a concussion near the beginning of the second quarter. His concussion occurred when he was hit and driven backwards into the ground striking the back of his head off the ground with a whiplash mechanism. He remained in the game and reported no issues; therefore, no medical evaluation was performed at the actual time of injury.

He had played in 22 plays prior to the concussion, with a performance grade of 67/100. After the concussion, he played in 27 plays and had a decline in performance grade by 24 points to 43/100, a 36% decline from his initial score (See Figure 1). In his one other game with significant playing time that year, he scored a 77/100 and did not demonstrate decrease in performance during that game.

One week after the game, the team reported back to the medical staff for the first time which is also the first time that he disclosed his concussion and symptoms. When he finally admitted to the injury and completed a SCAT-3 test one week after the injury, he endorsed 17 of 22 symptoms (baseline 2 of 22) with a severity of 42/132 (baseline 2/132). His SAC Score was 21/30 (baseline 29/30), with a Balance Score of 22/30 (baseline 4/30), and a Maddock's Score of 5/6 (Table 2). After returning to baseline four weeks after the concussion, he completed a return to play protocol over about two weeks and was fully cleared six weeks after the injury.

<Insert Table 2 here>.

DISCUSSION

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Our study found that college football players' PG were negatively impacted by a concussion and fell a substantial percentage off of pre-concussion scores. The numbers reported in these three cases are large enough that players and coaches should take note that concussions have large effects on in-game performance and should not be "played through." Performance grading in college football has both subjective and objective components. Objectively, the player can be evaluated on whether they line up in the correct place for a given play. Subjectively, the player can be evaluated on whether they make the correct read and subsequent action. Each player in a position group will have the same coach grading them, but coaches for different position groups may have slightly different grading systems. Since the same coach was grading the player both before and after the concussion, the scoring system should be the same which would provide an internal control. Furthermore, the scores are also checked in detail by Graduate Assistants and to some extent by the Head Coach and either the Offensive or Defensive Coordinator.

The raw change in performance grades for all three cases declined as did the calculated percent changes as a ration of their pre-concussion grades. We used a percentage of the pre-concussion grade in an attempt to control for external factors such as prior injury and quality of opponent, amongst others, which could affect the score. While a drop in raw score is interesting, it lacks meaning without knowing what the player's score was before the injury. For example, if player A scores 80/100 pre-concussion then 60/100 after a concussion, they are dropping 20 points, but when described as a ratio of their initial score, they are dropping 25% from initial score. This 25% is actually expressing the number of plays they will now make incorrectly that they would have made correctly before the injury. Coaches and players will find this number much more intuitive when they are told that a concussed player may miss 1 in 4 plays that they

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would normally make.The PG for an individual player may be quite different from game to game since the score is directly affected by the quality of the opponent as well as other game time variables such as injuries, playing conditions and game situations. This is why our primary outcome measure was comparing the PG before and after the concussion inside the same game. By focusing on the scores inside the same game, we effectively control for many of these variables. As a scoring system developed and utilized only by football coaches, there is no current published research using PG, and there are no studies of validity. In each of these three cases, the coaches had already completed their grading when approached by the authors. This was done deliberately to avoid bias when the coaches were grading performance.

For future prospective research with performance grades, it will be important to note if the player typically has poor grades later in games since it may just be the player's fatigue that is leading to a decrease in performance. We were unable to gain access to detailed reports of play by play grades for each player in every game of the season but attempted to get comments from each player's position coach to address this. It is reasonable to assume that the fatigue factor will be greater for larger players such as offensive and defensive lineman, but there are many variables that would impact this such as game-day weather, number of plays and quality of opponent.

Much of concussion research currently focuses on detection since we know that players can hide their symptoms well [32-34]. We continue to educate players on the long-term effects of concussions to get them to disclose symptoms of which we may be otherwise unaware [35,36]. Players are still reluctant to report concussions due to not wanting to be removed from current or future games, not thinking it was serious, and not wanting to let down their teammates [18]. While most medical providers would logically assume that performance is negatively

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affected by a concussion, athletes and coaches may not agree with this assumption. In fact, as mentioned earlier, the body of literature on game performance after a SRC shows no difference which may be misinterpreted by athletes and coaches since it is only looking at games after recovery from SRC [26-30]. We do not know of any published work looking at in-game decline in performance such as ours. We are seeking to present evidence of decreased performance which should strike a chord with both athletes and coaches. If players and coaches know that players may be hurting the team by making the wrong reads and missing assignments when playing with a concussion, players may actually be more apt to report symptoms, and coaches may be more forthcoming when they believe a player isn't acting normally. Additionally, we reported cases of football players hiding concussions, but we know that it happens in all other contact sports as well and would warn athletes of any sport that their performance may likely decline after a concussion to a level that is detrimental to the team.

With only three cases, the applicability of this work is limited and should be pursued further with a more rigorous research methodology in the future. Specifically, Case 2 has a very small sample of plays after the injury, and the position coach may have been biased by his own assessment of the health of the player. The fatigue factor as described above was not completely addressed as we were unable to get a more complete set of performance grades. There is a potential bias toward a larger effect on PG since two of the three cases self-reported symptoms after the game and since Case 2 must have displayed something abnormal which alarmed his position coach. It is logical to assume that there may be many other injuries that occurred to other players during each of these games that did not rise to the same level of these three cases which is the ultimate purpose of this case series to show that performance is affected when concussion symptoms are hidden, and in turn, this would encourage more athletes to come

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forward with symptoms when these injuries are not apparent to the medical staff. Finally, the performance grading scale is not a scientifically validated scoring system but conceivably has enough checks and balances to be useful. Performance grading may not be able to sufficiently compare one player to another with respect to medical research standards, but it would seem to be much more useful and reliable when comparing the scores of one player to himself over time as we have done here.

Conclusions

In this case series, we reported three college football players who suffered concussions, but they did not immediately disclose their symptoms. All three had a marked decrease in their performance grades post-concussion. This case series should serve as a first step in quantifying concussion effects on performance when athletes choose to conceal their injury. If the effect of a concussion on performance can be quantified and the potential detriment to the individual and team can be effectively emphasized, athletes and coaches may begin to change their mindset that a concussion can be "played through."

REFERENCES

[1] Zuckerman SL, Kerr ZY, Yengo-Kahn A, et al. Epidemiology of sports-related concussion in NCAA athletes from 2009-2010 to 2013-2014: Incidence, recurrence, and mechanisms. Am J Sports Med. 2015 Nov;43(11):2654-62.

[2] Tjong VK, Baker HP, Cogan CJ. Concussions in NCAA Varsity Football Athletes: A Qualitative Investigation of Player Perception and Return to Sport. J Am Acad Orthop Surg Glob Res Rev. 2017 Nov 20;1(8):e070.

ACCEPTED MANUSCRIPT

[3] Kerr ZY, Register-Mihalik JK, Marshall SW, et al. Disclosure and non-disclosure of concussion and concussion symptoms in athletes: review and application of the socio-ecological framework. Brain Inj. 2014;28(8):1009-21.

[4] Beverly EA, Fredricks, TR, Leubitz A, et al. What can family medicine providers learn about concussion non-disclosure from former collegiate athletes? BMC Fam Pract. 2018 Jul 27;19(1):128.

[5] Manley G, Gardner AJ, Schneider KJ, et al. A systematic review of potential long-term effects of sport-related concussion. Br J Sports Med. 2017 Jun;51(12):969-977.

[6] McCrea M, Guskiewicz K, Randolph C, et al. Effects of a Symptom-Free Waiting Period on Clinical Outcome and Risk of Reinjury After Sport-Related Concussion. Neurosurgery. 2009;65:876-83.

[7] McCrea M, Guskiewicz KM, Marshall SW, et al. Acute effects and recovery time following concussion in collegiate football players: the NCAA Concussion Study. JAMA. 2003 Nov 19;290(19):2556-63.

[8] Eckner JT, Kutcher JS, Richardson JK. Effect of concussion on clinically measured reaction time in 9 NCAA division I collegiate athletes: a preliminary study. PM R. 2011 Mar;3(3):212-8.

[9] Brooks MA, Peterson K, Biese K, et al. Concussion Increases Odds of Sustaining a Lower Extremity Musculoskeletal Injury After Return to Play Among Collegiate Athletes. Am J Sports Med. 2016 Mar;44(3):742-7.

ACCEPTED MANUSCRIPT

[10] Fino PC, Becker LN, Fino NF, et al. Effects of recent concussion and injury history on instantaneous relative risk of lower extremity injury in Division I collegiate athletes. Clin J Sports Med. 2019 May;29(3):218-23.

[11] Brooks MA, Peterson K, Biese K, et al. Concussion increases odds of sustaining a lower extremity musculoskeletal injury after return to play among collegiate athletes. Am J Sports Med. 2016;44(3):742-7.

[12] Herman DC, Jones D, Harrison A, et al. Concussion may increase the risk of subsequent lower extremity musculoskeletal injury in collegiate athletes. Sports Med. 2017;47:2003-10.

[13] McPherson AL, Nagai T, Webster KE, et al. Musculoskeletal injury risk after sport-related concussion. Am J Sports Med. 2019;47(7):1754-62.

[14] Howell DR, Wilson JC, Brilliant AN, et al. Objective clinical tests of dual-task dynamic postural control in youth athletes with concussion. J Sci Med Sport. 2019 May;22(5):521-25.

[15] Eagle SR, Kontos AP, Pepping GJ, et al. Increased Risk of Musculoskeletal Injury Following Sport-Related Concussion: A Perception-Action Coupling Approach. Sports Med. 2020;50:15-23.

[16] Asken BM, Bauer RM, Guskiewicz KM, et al. Immediate Removal From Activity After Sport-Related Concussion Is Associated With Shorter Clinical Recovery and Less Severe Symptoms in Collegiate Student-Athletes. Am J Sports Med. 2018;46(6):1465-74.

[17] Torres DM, Galetta KM, Phillips HW, et al. Sports related concussion: Anonymous survey of a collegiate cohort. Neurol Clin Pract. 2013 Aug;3(4):279-287.

ACCEPTED MANUSCRIPT

[18] Llewellyn T, Burdette GT, Joyner AB, et al. Concussion reporting rates at the conclusion of an intercollegiate athletic career. Clin J Sports Med. 2014 Jan;24(1):76-9.

[19] Conway FN, Domingues M, Monaco R, et al. Concussion symptom underreporting among incoming national collegiate athletic association division I college athletes. Clin J Sport Med. 2018. doi: 10.1097/JSM.0000000000000557. [Epub ahead of print].

[20] Delaney JS, Lamfookon C, Bloom G, et al. Why university athletes choose not to reveal their concussion symptoms during a practice or game. Clin J Sport Med. 2015;25(2):113-25.

[21] Craig DI, Lininger MR, Vomacka MM, et al. Concussion reporting behaviors of athletes: A systematic review. Ath Training and Sports Health Care. 2019. doi:10.3928/19425864-20190322-01.

[22] Kroshus E, Chrisman SP, Milroy JJ, et al. History of concussion diagnosis, differences in concussion reporting behavior, and self-described reasons for non-report. Jour of Clin Sport Psych. 2018. doi:10.1123/jcsp.2017-0036.

[23] Milroy JJ, Hebard S, Kroshus E, et al. Sport-related concussion reporting and coach-athlete attachment among collegiate student-athletes. Jour of Clin Sport Psych. 2018;12(2):268-81.

[24] Baugh CM, Meehan WP III, Kroshus E, et al. College football players less likely to report concussion and other injuries with increased injury accumulation. J of neurotrauma. 2019. doi:10.1089/neu.2018.6161

[25] Craig DI, Lininger MR, Wayment HA, et al. Investigation of strategies to improve concussion reporting in American football. Res in sports med. 2019. doi:10.1080/15438627.2019.1586706

ACCEPTED MANUSCRIPT

[26] Zuckerman SL, Kuhn AW, Gentry W, et al. Player Performance After Returning From a Concussion in the National Football League: A Pilot Study. J Surg Orthop Adv. 2018 Fall;27(3):187-197.

[27] Yengo-Kahn AM, Zuckerman SL, Stotts J, et al. Performance following a first professional concussion among National Basketball Association players. Phys Sportsmed. 2016 Sep;44(3):297-303.

[28] Van Pelt KL, Lapointe AP, Galdys MC, et al. Evaluating Performance of National Hockey League Players After a Concussion Versus Lower Body Injury. J Athl Train. 2019;54(5):534-40.

[29] Buckley TA, Bryk KN, Van Pelt KL, et al. Concussion and National Hockey League Player Performance: An Advanced Hockey Metric Analysis. J Athl Train. 2019;54(5):527-33.

[30] Kuhn AW, Zuckerman SL, Totten D, et al. Performance and Style of Play After Returning From Concussion in the National Hockey League. Am J Sports Med. 2016;44(8):2152-57.

[31] Zimmer A, Marcinak J, Hibyan S, et al. Normative values of major SCAT2 and SCAT3 components for a college athlete population. Appl Neuropsychol Adult. 2015;22(2):132-40.

[32] Oberlander TJ, Olson BL, Weidauer L. Test-Retest Reliability of the King-Devick Test in an Adolescent Population. J Athl Train. 2017 May;52(5):439-445.

[33] Patricios J, Fuller GW, Ellenbogen R, et al. What are the critical elements of sideline screening that can be used to establish the diagnosis of concussion? A systematic review. 2017 Jun;51(11):888-894.

[34] Mathiasen R, Hogrefe C, Harland K, et al. Longitudinal Improvement in Balance Error Scoring System Scores among NCAA Division-I Football Athletes. 2018 Feb 15;35(4):691-694.

ACCEPTED MANUSCRIPT

[35] Kroshus E, Daneshvar DH, Baugh CM, et al. NCAA concussion education in ice hockey: An ineffective mandate. Br J Sports Med. 2014 Jan;48(2):135-40.

[36] Register-Mihalik JK, Guskiewicz KM, Valovich McLeod TC, et al. Knowledge, attitude, and concussion-reporting behaviors among high school athletes: A preliminary study. J Athl Train. 2013 Sep-Oct;48(5):645-53.

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FIGURE LEGENDS

Figure 1. Raw performance grades for all three cases, comparing pre- and post-concussion.

Figure 1

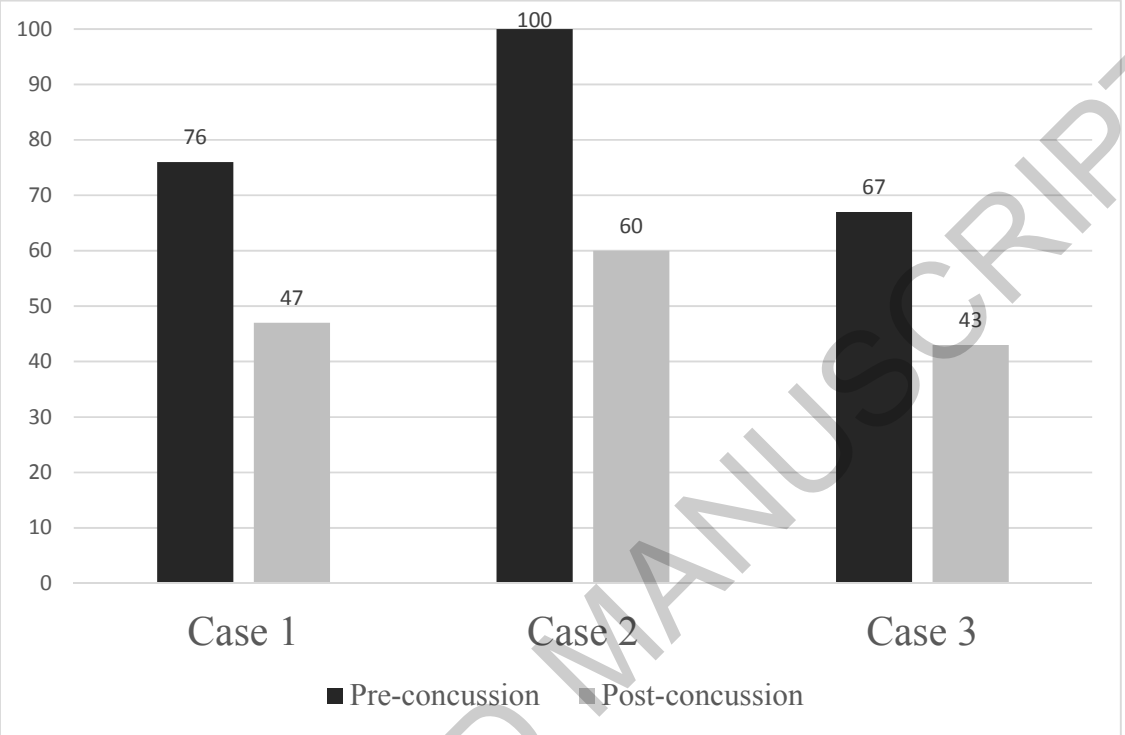

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Table 1. Basic demographic data for study subjects

	Case 1	Case 2	Case 3
Age	21	21	20
Ethnicity	African-American	African-American	Caucasian
Position	Defensive line	Defensive skill	Offensive skill
Prior concussions (time from current concussion)	2 (2 months, 14 months)	1 (10 years)	0
Game time of concussion	3rd quarter	1st quarter	2nd quarter
Plays before concussion	38	19	22
Plays after concussion	17	5	27
Approximate time to presentation	1 hour	10 minutes	1 week

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Table 2. Initial post-concussion evaluation scores.

	Case 1		Case 2		Case 3	
	Baseline	**Post-injury**	**Baseline**	**Post-Injury**	**Baseline**	**Post-Injury**
Symptom Score (0-22 where 0 is best)	6	18	1	1	2	17
Symptom Severity (0-132 where 0 is best)	12	64	4	2	2	42
Worse with physical activity?	No	No	No	No	No	Yes
Worse with mental activity?	No	Yes	No	No	No	Yes
Standardized Assessment of Concussion Score (0-30 where 30 is best)	28	26	26	25	29	21
Balance Score (0-30 where 0 is best)	5	7	5	2	4	22
Maddock's Score (0-6 where 6 is best)	NA	5	NA	5	NA	5

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